

## QUESTION 1

# Preparing a Greenhouse Gas Inventory

---

What to count, where to draw the line, and what the answer changes.

Worked on a 120,000 barrel-per-day Nigerian refinery

GROUP C (29 MEMBERS)

# Three scopes. One fence.

The scopes exist so that two companies never count the same tonne twice.

## SCOPE 1

Direct

Everything burned, flared, vented or leaked inside your fence.

## SCOPE 2

Indirect — energy

The power station that runs your lights.

## SCOPE 3

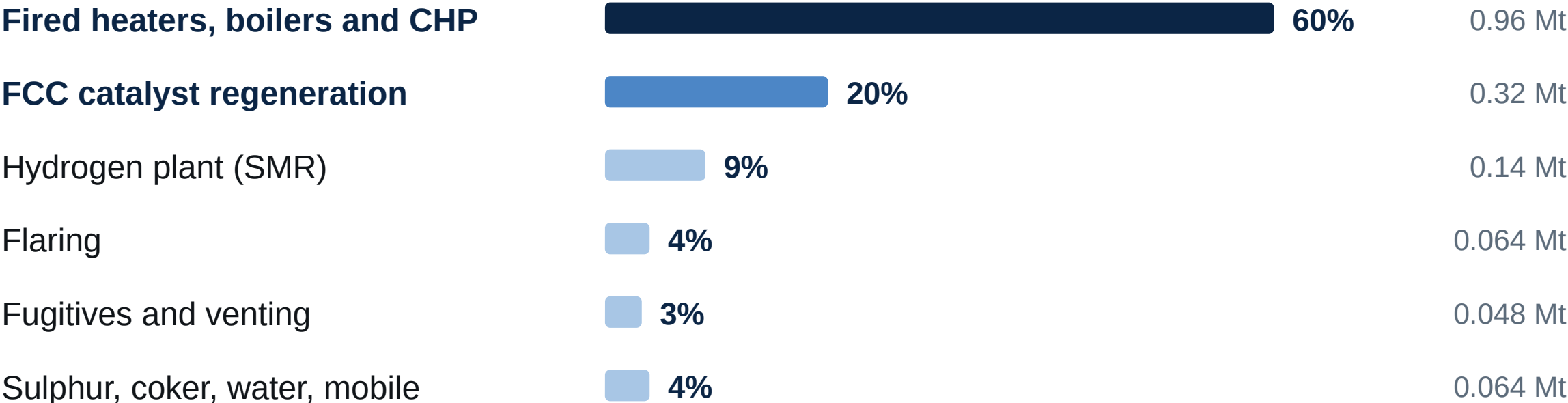
Indirect — everything else

Your crude supplier, and every vehicle burning the diesel you sold.

**Scope 1 and 2 are what you control. Scope 3 is what you cause.**

# Two sources carry four fifths of it.

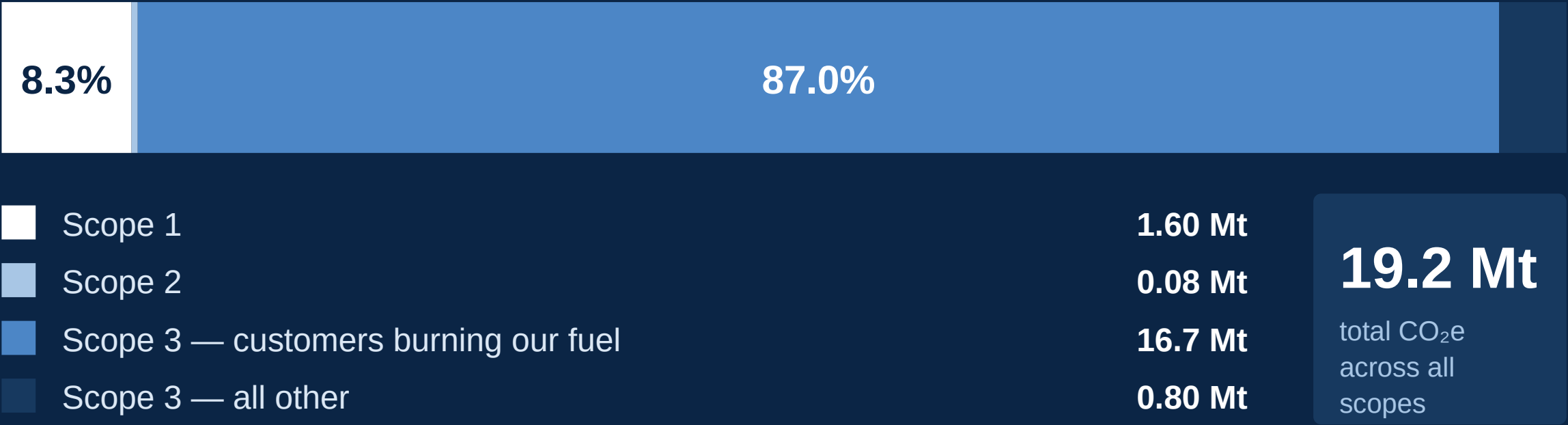
A refinery is not one chimney. It is about a dozen distinct mechanisms.



**The two most often missed:** the FCC burns no purchased fuel, and the hydrogen plant releases CO<sub>2</sub> from the reaction, not the flame.

THE NUMBER THAT CHANGES THE CONVERSATION

87% of the footprint leaves the gate in a tanker.



Every efficiency lever works on the 8.8%. Moving the 87% is a question about what we sell.

# Measure it well, then act on it.

An inventory that changes no decision was a bookkeeping exercise.

HOW WE MEASURE

- 1 Continuous monitoring in the stack
- 2 Mass balance — carbon in, carbon out
- 3 Engineering calculation
- 4 Published emission factor

*Best evidence at the top. Say which you used.*

WHAT WE DO ABOUT IT

|                               |      |
|-------------------------------|------|
| Energy and heat integration   | 3–8% |
| Methane leak detection        | 1–3% |
| Flare gas recovery            | 2–4% |
| Capture on the hydrogen plant | 7–9% |

*Cut to Scope 1, cheapest first.*

**An inventory is a strategy document that happens to be written in tonnes.**